

Adaptive Prompt Compiler: A Headroom Typology for Prompt Optimization on Real Reasoning Models (Anonymous Preprint v0.3)

Anonymous submission

Abstract

APC turns prompts from hand-written text into compilable engineering objects: a task specification (TaskSpec) compiles, together with an explicit 15-dimensional model profile (ModelProfile), through a discrete 10-gene genome (PromptGenome) into a prompt, which is then improved by budget-aware evolutionary search (with a profile-guided mutation operator, PGAM) and carried across models by structural migration. On a deterministic four-task benchmark (APCBench; financial/contract/math/constraint-following $\times$ 3 simulated model profiles): search significantly beats zero-shot (+0.009 to +0.029); evolution beats random search on compositional tasks (+0.003) but loses on single-locus tasks (-0.013, successive-halving selection noise); PGAM is indistinguishable from uniform mutation (pooled +0.0004, a null result; its first real-model test, gpt-6 math, runs -.0376 vs uniform); migration reaches native-quality at 30% of the re-search budget (KR-6 passed 12/12). On a real frontier reasoning model (qwen3.8-flash, four arms $\times$ three tasks, identical rule-judge): accuracy gains of genome search are ≈ 0 — the base-root arm holds the zero-shot floor on all three tasks (within -0.004) while the profile-rule root is a net liability (dev-score deficits of 0.2–0.53, recovered only partially within budget); the strongest deployment policy is *zero-adaptation transfer* of a cross-task champion genome (0 budget, $\geq$ same-task zero-shot, up to +0.79 over cold re-search). External author-unseen benchmarks (GSM8K, MATH Level-5, AIME 2024+25) reproduce both findings: all six champion-vs-base deltas are within noise and the model saturates every benchmark (AIME: 60/60), bounding the room for prompt optimization on strong reasoners. We distill these into a methodological principle, *Compile-as-Hypothesis*: compiled artifacts are testable hypotheses, never default deployables, and prior value is a measured quantity that can be negative. The multi-model repli-

cation matrix ($\backslash S\}$ 3.4p) confirms the task-regime taxonomy and cross-task warm-start migration on two additional frontier reasoning models (gpt-6, gpt-5.6-terra via an OpenAI-compatible gateway), with the seed-42 champion genome byte-identical across all three models. On the same real API we then close the baseline and regime questions: the official GEPA baseline (Algorithm 1 reproduced) is also within the paired same-day noise band (+.0024), the second strong baseline ESPO likewise (.7009), and a pre-registered token-starved discriminative experiment (F10) pins six pathways at the judge floor (.1500) when headroom is physically unreachable. F11 completes the map on a stratified 90-problem HLE-exact subset — the first REAL reachable knowledge-type headroom (zero-shot accuracy .10): all six arms tie again (.388-.412); a confirmatory $n=100$ round replicates the tie (gepa vs z0 McNemar $p=.219$) and measures the same-text nondeterminism floor at 2.7%, a GPQA-Diamond round ($n=190$) replicates the tie on a second real dataset (five arms .853-.868; reflective search accepts zero candidates), even though GEPA demonstrably learned domain content rules (SMILES conventions, perturbation-theory regime qualifiers). The discriminator for prompt-optimization payoff is not whether headroom exists but what kind: physical-truncation and knowledge-type headroom are out of reach for any prompt pathway (ability lives in weights), while protocol/format headroom is reachable and favors structured genomes — deployment should begin with a headroom-type diagnostic ($\backslash S\}$ 3.4m). A deployment gate (AutoAPC-Select, F8: val-argmax + noise-band Occam tie-break, 5/7 exact-oracle on 7 replay+blind groups) turns these findings into a selection rule. Multi-model validation of PGAM awaits additional API credentials.

1. Problem and Claim

The same business task requires different prompts on different LLMs (Sclar et al. 2023: format sen-

sitivity is model-specific; FLASK, Ye et al. 2024: skill profiles differ per model). Existing evaluation suites (HELM/FLASK) rank models but do not guide prompt construction; existing prompt compilers (DSPy) re-search per model but keep the profile implicit and uninterpretable (see docs/literature/lit-compiler.md G1–G6).

APC's claim: **an explicit capability profile (15 dims) as a compiler intermediate representation** — TaskSpec $\rightarrow$ PromptGenome (10 genes) $\rightarrow$ ModelProfile $\rightarrow$ PromptCompiler (rule compilation) $\rightarrow$ TrialResult $\rightarrow$ evolutionary search (PGAM) $\rightarrow$ model migration, fully traceable end-to-end (KR-3).

Contributions (ordered by reviewer concern, not chronology):

- **C1 · Profile-driven traceable compilation:** the TaskSpec $\times$ PromptGenome $\times$ ModelProfile IR attributes every prompt change to a gene locus and a profile dimension ($\S$ 2.1-2.3, F4 decision-chain audit) — a capability DSPy-style implicit profiles cannot offer.
- **C2 · Quantified liability of priors:** the rule-root costs $\sim .090$ on a real model (robust across three seeds, including a bimodal seed-collapse case); "compilation priors can be negative and are measurable" — Compile-as-Hypothesis becomes a quantity, not a slogan ($\S$ 3.4 F2/F7).
- **C3 · A complete discriminator for APO pay-off:** four independent pathways tie in the saturated regime (F7-F9) $\rightarrow$ all tie under physical truncation (F10) $\rightarrow$ all tie under reachable knowledge-type headroom, with mechanism evidence that reflection learned content rules yet transferred zero (F11) $\rightarrow$ while structured pathways win $+0.05$ in simulated protocol-type headroom. Merged into a **headroom taxonomy**: when no APO can win — prior to "whose representation is better" ($\S$ 3.4 F10/ $\S$ 3.4m).
- **C4 · Zero-cost deployment gate:** AutoAPC-Select spends measured same-day drift bands as its tie-break currency; on 7 replay+blind groups, 5/7 exact-oracle, both seed-collapse arms rescued at zero regret ($\S$ 3.4 F8).

- **C5 · Evaluation-methodology assets:** the hardened exact-match judge v3.2 (F6), the paired same-day re-measurement protocol (F7), two empirical instances of the val/holdout common-pool clause (F10/F11), and the connection-layer reliability triad for long-run harnesses (F11) — all in-repo.

2. Method

2.1 PromptGenome: a discrete structural genome

Ten gene classes (role/goal/instructions/constraints/examples/reasoning/verification/output/style/layout); search_space declares mutable loci (13 in our experiments). Unlike continuous soft prompts (Lester 2021; SPoT), a discrete genome is re-compilable and portable across models ($\S$ 3.3).

2.2 Profile-Guided Adaptive Mutation (PGAM)

- Prior: $w = 1 + 2(1 - \text{capability})$ truncated to $[0.5, 3.0]$; unmapped loci get weight 1.0.
- Online correction: per-generation surviving elite mutation loci $\times 1.3$ (cap 5.0), floor 0.3 — an ϵ -guarantee that lets the bandit override a mis-specified profile.
- The sole difference from uniform mutation is the locus sampling distribution; scheduling, budget accounting and lineage logging are identical, making the ablation a clean single-variable contrast.

2.3 Budget-aware population + Successive Halving (SHA)

$P = \min(P_0, \max(E+1, (B-3)/(G+2)))$; per generation dev_r1 halves the field and dev_full re-scores. Dev selects, validation names the champion, holdout runs the champion exactly once (the EvoPrompt discipline).

2.4 Migration protocol (FR-7)

CapabilityDelta (15 dims) $\rightarrow$ MigrationMutator adjusts the seed $\rightarrow$ small-budget re-search on target $\rightarrow$ adopt iff holdout $\geq 90\%$ of source (KR-6).

2.5 Design principle: Compile-as-Hypothesis

Real-model evidence ($\S$ 3.4 F2) forces the principle: **the output of profile-rule compilation is**

a hypothesis to be tested, not a deployable default. The pipeline therefore always runs two roots in parallel at equal budget — rule-root (apc-full) and base-root (apc-safe) — and deploys the winner. This makes the implicit assumption of DSPy-style per-model re-search explicit and measurable: prior value = $\Delta(\text{rule-root}, \text{base-root})$, and it can be negative (measured here: 2 of 3 tasks $\leq +0.001$; independently corroborated by the capability-dependent compilation gains of arXiv 2608.02639). The principle does not arrive *ex nihilo*: the spirit of deterministic acceptance layers over free-form judging is now established in production-systems literature (PROCTOR catalogues 11 evaluation- signal failure modes with a judge-overridable-proof acceptance gate, arXiv 2609.02246; accept-or-revert gate family: TARA arXiv 2607.18724, arXiv 2606.30840, SSO arXiv 2607.28777; a measurable-prior Bayesian cousin: Textual Bayes arXiv 2506.10060). Our contribution is to *name and operationalize* this practice as a compiler design principle — both roots permanently in the pipeline — and to attach quantitative deficit readings (F2/F7). The same principle governs migration: seeds must pass the holdout test before being carried (KR-6).

3. Experiments (APCBench, experiments/apcbench/)

- Task A "Financial Report Analysis" (dev 40 / validation 30 / holdout 30 / perturbation 30); Task B "Contract Information Extraction", Task C "Math Word Problems", and Task D "Verifiable Constraint Following" (all dev 40 / validation 30 / holdout 30; generators and seeds are in the respective task scripts). B/C share the `_prompt_skill` gene dynamics (different domain logic); D uses in-task dynamics (three channels: constraints / sentence count / numbers, activating the instructions/constraints loci that were neutral until now). Perturbation design (A only): distractor sentences + replacement of the first metric value (the gold standard is kept on the clean version).
- Models: glm / qwen / gpt (deterministic simulation via MockClient; ground truth in the README).

- Methods: zero-shot / manual / random-search / apc-full (uniform evolution + SHA) / apc-no-profile / apc-no-halving / apc-pgam.
- Statistics: Task A 5 seeds $\times$ 3 models, B/C/D 3 seeds $\times$ 3 models; paired bootstrap 95% CI.

3.1 Main results (b100, holdout)

Comparison	Δ	95% CI	Conclusion
apc-full - zero-shot	+0.0093	[0.0082, 0.0103]	significant; search works
apc-pgam - zero-shot	+0.0101	[0.0095, 0.0107]	significant
evolution - random	+0.0030	[0.0022, 0.0038]	significant; value of multi-step composition
apc-pgam - apc-full	+0.0008	[0.0000, 0.0017]	marginal ; variance 0.0001 vs 0.0016
apc-full - manual	-0.0009	[-0.0017, -0.0001]	a strong manual heuristic still leads
SHA / rules ablation	0.0000	—	null result (base already satisfies the rules)

3.2 Budget curves (holdout mean)

The plateau is reached at 50 evaluations (a consequence of the task's unimodality); random in fact degrades at b100 (single-step space + more candidates $\rightarrow$ dev overfitting).

3.2b Second task: contract extraction (b50, holdout, n=9/method)

3.2c Third task: math word problems (b50, holdout, n=9/method)

Comparison	Δ	95% CI	Conclusion
search - zero-shot	+0.0285	[0.0191, 0.0372]	significant
search - manual	-0.0109	[-0.0154, -0.0063]	significant; manual is strong (exact count/strategy is hand-optimal)
pgam - full	0.0000	—	no difference

Cross-task consistency: search $\gg$ zero-shot holds on all four tasks (+0.009 / +0.028 / +0.029 / +0.029); search vs manual: financial -0.001, contract +0.003 (n.s.), math -0.011, constraint following +0.024 (significant) — strong handcrafted

budget	pgam	full	random	manual	zero-shot
25	0.7501	0.7495	0.7497	0.7595	0.7493
50	0.7581	0.7581	0.7581	0.7595	0.7493
100	0.7600	0.7592	0.7561	0.7600	0.7499

Method	Mean	Comparison	Δ	95% CI
apc-full / apc-pgam	0.9844	search - zero-shot	+0.0275	[0.0130, 0.0420], significant
manual	0.9814	search - manual	+0.0031	[0.0000, 0.0061], not significant
zero-shot	0.9569	(qwen alone 0.9300 $\rightarrow$ search repairs it to 0.9833; weak-model rescue +0.05)		

heuristics are hard to beat on a unimodal simulator; the value of automation lies in no handwork, plus transfer, plus lineage.

3.2d Fourth task: verifiable constraint following (b50, holdout, n=9/method)

Comparison	Δ	95% CI	Conclusion
search - zero-shot	+0.0292	[0.0171, 0.0399]	significant
search - manual	+0.0238	[0.0116, 0.0341]	significant; search beats manual significantly for the first time
random - evolution	+0.0132	[0.0035, 0.0246]	significant; direction reversed (see below)
pgam - full	+0.0005	[-0.0148, 0.0162]	not significant

Optimizer cross-decomposition (after adding a no-halving control for Task D): no-halving (0.9616) > random (0.9513) > SHA evolution (0.938). no-halving - random: +0.0103 [0.0022, 0.0205] (significant) — when selection is accurate, the compositional value of multi-step evolution is positive; full - no-halving: -0.0235 [-0.0357, -0.0112] (significant) — SHA's small-sample dev_r1 screening comes at a heavy cost on this task. Conclusion: **the speed-accuracy trade-off is quantifiable, and there is no free lunch**; SHA favors compositionally hard tasks (financial), while full-population selection favors single-locus tasks (constraint following). PGAM is likewise indistinguishable from uniform on this task (+0.0005).

3.2e PGAM pooled across tasks (combined paired deltas, n=42)

pgam - uniform: +0.0004 [-0.0029, 0.0038] — a **null result** (the CI still crosses zero after narrowing). The profile prior delivers no measurable gain

on this simulator (the profile itself is weakly informative: the three models differ on only 3/15 dimensions; and the optimal basin is a single one). Reasons to keep PGAM: lower variance (b100 std 0.0001 vs 0.0016), an interpretable prior, and a bandit lower bound guaranteeing it is no worse; confirming or falsifying it requires genuinely multimodal real tasks.

3.2f Robustness (perturbation, champion genome, n=15/method)

All methods drop from holdout $\rightarrow$ perturbation by ≈ -0.010 (a numeric-substitution offset independent of the genome); there is no evidence that search champions are more fragile (the ranking is consistent with holdout).

3.3 Transfer (4 directions $\times$ 3 seeds, adapt budget 30 vs native 100)

- recover (adapted/native): 1.0000 [0.9976, 1.0025]; KR-6 pass rate 12/12; adapt - direct: +0.0001 (direct transfer is essentially lossless — the heterogeneity sits below the optimizer's resolution).
- Conclusion: the transfer protocol reaches native re-search quality at **30% of the budget** (a GEPA-style rollout argument), with lineage and an adopt/keep decision audit chain attached; the flatness of the decay matrix itself is a limitation of this simulator.

3.4 Real-LLM validation (phase one: qwen3.8-flash $\times$ 3 tasks, bench_real_full/transfer.py)

Setup: whitelisted-key model, reasoning model (~29 s/sample), temp=0; scoring uses the **same rule-judge** as the simulation benchmark; all three tasks share the identical dev_r/val/holdout = 5/8/20 split;

budget 8 for the main comparison, budget 6 for the transfer experiments.

- **F1 ceiling effect (double saturation of accuracy and robustness):** on a strong reasoning model, the accuracy headroom on rule-verifiable tasks is ≤ 0.005 , and the robustness headroom under input perturbation is likewise ≤ 0.013 (F5) — the preset stance that "the value of prompt optimization lies in the format/constraint dimension" is **falsified on the real model**; the only positive value remaining is weak-prior rescue (F2, with no guarantee of full repair) and zero-loss transfer (F3).
- **F2 the rule prior is the dominant risk on real models; base-root search is a "lossless floor", rule-root search is "high-cost rescue":** the dev scores of the profile-compiled root are contract 0.1778 / math 0.6335 / financial 0.1286 — all 0.2–0.53 below zero-shot. The b8 search pulls the rule-root arm back into the examples-on basin, but **repair completeness decreases as task headroom grows**: contract 0.9782 $>$ z0, math 0.8442 $\approx$ z0, financial 0.5704 still -0.101; at budget 6 the search never even samples the repair site (cold 0.1850). The control arm **apc-safe (base root + the same b8 budget) lands within -0.004 of zero-shot on all three tasks** (0.9773/0.8436/0.6672): after spending the same budget, the search neither gains nor loses, and the financial champion is base + goal.explicitness: high $\rightarrow$ low (val selection noise, holdout -0.004). Conclusion: **on real models the profile rule prior is a net liability; structural genome search yields accuracy gains $\approx$ 0, and its only value is rescuing a bad prior (without guaranteeing a full repair)**. The simulator's "apc-full $\equiv$ zero-shot" safety does not carry over to real models — the source of unsafety is precisely the compiled rules.
- **F3 cross-task genome transfer (two directions, b6 same-budget three arms):**

Transfer	cold (search)	transfer-0 (zero-adapt direct use)	transfer-ws (continued search)
contract $\rightarrow$ math	0.6402	0.8439	0.8441
math $\rightarrow$ contract	0.1850	0.9778	0.9582

The strongest arm is **zero-adapt direct use of the source champion** (transfer-0), reaching the

target task's zero-shot level at 0 budget; warm-start continued search yields ≈ 0 gain on saturated tasks (math +0.0002) or even negative gain (contract -0.0196 — val_n=8 selection noise picks an overfit champion); cold start lags significantly because it must first pay F2's repair cost. Mechanism: the genome carries full structure while few-shot content is injected from the TaskSpec at compile time $\Rightarrow$ zero-loss across tasks; the optimal basin is unique (both tasks' champions are examples-on), cross-validating the simulator's "single-peak" finding. **Engineering implication: deployment order transfer-0 $\rightarrow$ base/apc-safe $\rightarrow$ rule-root with budget $\geq$ repair cost; never default to a cold-start re-search.**

- **F4 judge-disagreement audit (20 cases from the financial champion, self-LLM-judge, real_judge_check.py):** on the accuracy dimension, rule mean 0.206 vs LLM 0.830 (Pearson 0.48 / Spearman 0.56; 20/20 disagree by ≥ 0.25); on the constraint dimension the direction reverses: rule constantly 1.0 vs LLM 0.50. Reading: the rule-judge is a gold-standard tolerance check, systematically stricter than the self-evaluating LLM, and the two judges are strict along *different* dimensions. $\Rightarrow$ All real-LLM ranking conclusions in this paper are valid under the same-protocol rule-judge, but **absolute scores must not be read as "human quality scores"**; an independent third-party judge (not self-evaluation) remains a gap. The self-judge's leniency may also contain a model self-preference bias.
- **F5 perturbation-robustness audit (first 20 cases of the financial number-tampering set $\times$ 4 genomes, bench_real_robust.py):** pert scores base 0.6580 (drop +0.0132) / manual 0.6654 (+0.0008) / apc-full 0.5820 (-0.0116) / apc-safe 0.6582 (+0.0090). All $|\text{drop}| \leq 0.013$ — strong models are themselves insensitive to numeric-tampering perturbations, and format genes have no realizable robustness premium (apc-full's negative drop is its overall level shift falling into the perturbation-set noise, not "more robust"). $\Rightarrow$ Merged with F1: **on real strong models, structural genomes are saturated in both accuracy and robustness; the gain budget can only come from rescue and transfer, not from optimization itself.**
- **F6 external real benchmarks (GSM8K /**

Task	zero-shot	manual	apc-full(rule-root)	apc-safe(base-root)
contract	0.9770	0.9773	0.9782	0.9773
math	0.8436	0.8439	0.8442	0.8436
financial	0.6712	0.6662	0.5704	0.6672

MATH Level-5 / AIME 2024+25, bench_real_external.py, exact-match judge with latex structural normalization, zero-adapt three arms): F1/F3 tested directly on tasks the authors did not build — genomes compiled via the external `external_math.yaml` spec; arms = base / math-champion (source task) / contract-champion (cross-domain), all evaluated on datasets that never participated in any genome.

Dataset (n)	base	math-champ (transfer-0)	contract-champ (cross-domain)
GSM8K (100)	0.96	0.98	0.98
MATH-L5 (135)	0.9704	0.9481 (-1.0 σ)	0.9704
AIME 24+25 (60)	1.00	1.00	1.00

After offline re-judging under judge v3.2 (exact-match + latex structural normalization): none of the six champion-vs-base comparisons is significant, and **on AIME 2024+25 all three arms score 60/60** (a frontier reasoning model gets a perfect olympiad score $\Rightarrow$ the external benchmarks fail even at being "hard"); the cross-domain champion (contract $\rightarrow$ math) is lossless just like the source-domain champion. The only consistent directional signal: math-champ is -0.012~-0.022 on both math sets (pooled -1.1 σ , not significant) — a **faint echo of the prior liability on external tasks**, same direction as F2 but two orders of magnitude smaller (the genome has already evolved into a near-base form). Judge reliability: 12 failed samples manually audited $\rightarrow$ three repair rounds v2 $\rightarrow$ v3.1 $\rightarrow$ v3.2 (matrices / solution sets / units / leading-zero flattening + addition-commutativity sign-term multiset; policy: sacrifice the strictness that (1,2) $\neq$ (2,1) in exchange for equivalence of all notation variants, and judge as wrong when in doubt) $\rightarrow$ 27 positive + 4 negative regression cases + full predictions persisted for

- Honest boundary: the external benchmark arms are single-model single-seed (no CI);

the multi-model matrix (`\S{}`3.4p) extends the core arms (financial/contract/transfer) to gpt-6 and gpt-5.6-terra, with the PGAM real validation run on the discriminative math cell (single seed).

F7 GEPA official-baseline head-to-head + variance audit (gepa vs full/safe/z0, bench_real_gepa.py / bench_real_reeval.py)

Control design: we reproduce the core of GEPA (Agrawal et al. 2025, arXiv 2507.19457) Algorithm 1 — failure-driven reflective rewriting from a minibatch (3) (free-text mutation), parent sampling from the per-instance Pareto frontier, and full-val re-evaluation to select the champion; identical to APC in starting point (the compiled z0 text), judge path (TrialScorer), budget currency (48 task roll-outs $\approx$ APC b8's actual spend of 25–35), and final holdout20 testing. Reflective LLM calls are not charged to the budget (the same logic as the APC compiler's overhead).

The variance-audit finding precedes every comparison: **the reasoning model is non-deterministic at temp=0** — the same z0 prompt scores .6950/.6997/.6998/.7016 on four same-day holdout20 re-measures (full range .0066), and drifts across days (yesterday .6712 $\rightarrow$ today .6990, Δ .028). Therefore **any cross-day comparison of numbers is invalid**; every conclusion in this section rests on **paired same-day re-evaluation** (the reeval batch, seed 902).

Same-day paired four arms (financial, completed within one hour window):

Method (same-day re-measured)	holdout	vs z0
z0 (same-batch ref; 4 same-day measures: .6950/.6983/.6998/.7016)	.6983	—
manual	.7012	+ .0029 (noise band; last night's .6662 was a low-drift day)
APC-safe champion	.7008	+ .0025 (within the noise band)
GEPA champion	.7014	+ .0024 (within the noise band)
APC-full champion	.6082	-.090 (a real liability)

Seed stability of the searches (each seed carries its own within-day baseline, hence naturally paired): APC-full across four seeds = .5704/.5705/.1334/.1318 — a **clean bimodal distribution** (a $\sim .57$ deficit basin and a $\sim .13$ collapse basin, 2/4 collapse rate); collapsed arms show the failure in val too (.1452/.1708, visible to any selector) — **rule-root mutation exhibits seed fragility**: beyond the deficit (-.09) there is bimodal risk. APC-safe's three seeds .6672/.6679/.6575 (full range .010, same magnitude as the day-drift) are stably indistinguishable. GEPA on contract (generic-judge caliber — **not comparable with APC's contract case caliber**): .7491/.7495 vs the same-caliber z0 .7500 — on a saturated task there is no failure signal to reflect on, zero progress, corroborating evidence for the null.

Three conclusions: (i) **the strongest open-source baseline GEPA also achieves zero gain on the strong reasoning model** (+.0024, within the noise band) — the $\set{S}{3.4}$ null is not a local defect of APC's representation but a property of the "strong model + high judge-coverage" regime; GEPA's reflection path and APC's structured-mutation path flatline alike here. (ii) APC's differentiated value holds beyond the null: the auditable decision trace (F4), zero-cost transfer (contract $\rightarrow$ math champion .9808), and the schema/robustness machinery (F3, the hardened external judges of F6) — none of which free-text mutation provides. (iii) **Methodological action**: we adopt the established practice of same-day test-retest floors and matched controls (arXiv 2608.00705; arXiv 2608.08239, COLM 2026; temp-0 nondeterminism quantified in arXiv 2408.04667, 2606.26185; reporting standards in arXiv 2607.24372) and institutionalize it for APO arm comparisons: to our knowledge the first ex-

plicit variance decomposition (between-day .028 vs within-day $\pm .007$) for reasoning-model prompt optimization, from which we draw the protocol clause that any arm-level claim within $\pm .01$ must rest on paired same-day re-measurement — indistinguishability inside the noise band being an information-theoretic necessity under verification lower bounds (arXiv 2604.12951). Our main four-arm table was run in one night under one protocol; the GEPA/reeval batches are paired same-day supplementary measures.

F8 AutoAPC-Select: a deployment-time gating selector (`auto_apc_gate.py`, offline replay, zero additional rollouts). F7's two negative findings (rule-root liability + seed fragility) yield a constructive corollary: per-arm validation scores are free on any given day, so deployment selection can be *automatically gated* — argmax over candidates {z0, manual, safe, full, transfer} on validation score, with ties inside the $\pm .007$ noise band broken by Occam order (z0 < manual < safe < full). Offline replay over the 12 existing arm groups plus a prospective blind-test group: 6/7 land within the noise band, 5/7 are exact oracles, and both collapsed rule-root arms (s44 $\rightarrow$.1334, s45 $\rightarrow$.1318) are fully rescued (gate picks safe .6575 / z0 .6692, regret 0). The s45 group is **prospective**: the gate rule (val argmax + $\pm .007$ Occam tie-break) was frozen before the group ran, and the gate's pick equalled that day's oracle — evidence, not post-hoc fitting. The single large regret (math $\rightarrow$ contract, +.0196) exposes the gate's failure mode: val(8)/holdout(20) distribution mismatch can inflate a candidate's validation score (transfer vs val .9822 > transfer-0 .8871 while holdout reverses, .9582 < .9778); the honest corollary is to bias noise-band ties toward the simpler arm or to draw val and holdout from a common pool. This upgrades F7's manual two-arm contrast into an automatic non-inferiority selector with a characterized failure mode — at zero extra evaluation cost, reusing existing validation scores. Boundaries within the gating family: ESPO (arXiv 2609.04197, EMNLP 2026 main) buys selection stability with bootstrap re-sampling (extra evaluation budget); AutoAPC buys it with the *measured exogenous drift band* at zero extra calls, requiring only that same-day val scores exist. Pure-exploration bandits (arXiv 2605.14553, ICLR 2026) give best-feasible identification with sampling theory but consume budget; AutoAPC is its free-score degenerate case. PROCTOR's deterministic acceptance layer guards

against judge hacking — orthogonal to, and stackable with, APC's variance guard (APC's own judges are deterministic checkers). Indistinguishability inside the band is not an engineering compromise: verification-tax lower bounds (arXiv 2604.12951) show 23% of frontier-model pairwise comparisons are statistically unresolvable in principle.

F9 ESPO, the second strong baseline (bench_real_espo.py, faithful reproduction of EMNLP 2026 main's Diagnose/Propose/Select). ESPO is currently the strongest published GEPA successor (its pitch: fix GEPA's prompt bloat and unreliable selection; +3.76pp over GEPA on 7 benchmarks). We reproduce its three steps: Diagnose clusters ALL validation failures into at most 4 error patterns in one pass (vs GEPA's incremental 3-case minibatch reflection); Propose draws one candidate from each of four independent bias strategies (root-cause abstraction / simplification / exemplification / constraint hardening); Select accepts a candidate only if it beats the incumbent in $\geq 75\%$ of $B=200$ bootstrap resamples of per-case validation scores. Budget matches the GEPA arm (48 rollouts, 40 used); bootstrap costs zero additional calls. Paired same-day result (seed-42 batch, z0 day band .6950-.7016): **ESPO .7009, inside the band and statistically indistinguishable from z0/safe/GEPA** — the fourth independent optimizer family (reflective GEPA, cluster-and-stabilize ESPO, APC genome search, manual) all null in the same regime. A second seed scored .7501 but its champion text was not persisted, there is no same-day z0 pair, and it crossed a day boundary — under the F7 protocol clause it is excluded from conclusions and disclosed as a live case of why unpaired numbers are untrustworthy. Relationship to our F8 gate: ESPO's stability currency is endogenous (re-sampling its own evaluations); APC's is exogenous (a measured drift band, at zero extra cost) — orthogonal and stackable, and this run illustrates exactly that distinction.

F10 Pre-registered discriminative experiment: the token-starved regime. Before running it we committed (git e8bd373, 22:20) a three-branch prediction: capping output at max_tokens=150 collapses zero-shot holdout from .699 to .150, opening apparent headroom; the question was whether genome search, free-text reflection (GEPA), and cluster-and-stabilize selection (ESPO) diverge once the signal is winnable. Result — branch (c), the pre-registered null-of-the-null: all six pathways

measured .1500 on a paired same day (z0 / manual / APC-safe / APC-full / GEPA / ESPO). Truncation occurs mid-JSON (verified in fail samples); the floor is the judge's format-collapse value, not model competence — **no prompt rewrite stops a reasoning model from spending its output budget** (instruction-level thought suppression is ineffective on this model). F10 does not overturn F1-F9; it completes the regime map: optimizer representations are equivalent-to-zero whenever headroom is either absent (F1-F9) or physically unreachable by prompting (F10); where headroom is genuinely reachable, the structured pathway wins (simulator weak-model rescue, +0.05, \S{}3.2b). Side finding for evaluation protocol: GEPA scored .6961 on the starved val-8 (short documents finish inside 150 tokens) while the starved holdout-20 scored .15 — truncation-difficulty asymmetry between a small validation set and holdout makes reflective optimizers adapt to the wrong regime; this is a concrete instance of the val/holdout common-pool clause of F8's failure-mode analysis.

F11 Headroom taxonomy: the HLE-exact high-difficulty contest (pre-registered P1 branch hit). bench_real_hle.py; pre-registration docs/literature/F11-prereg.md (git 262a39e, locked before any gepa/espo holdout number existed); dataset generator scripts/gen_hle_dataset.py.

Motivation: after F1-F9 (no headroom) and F10 (physically unreachable headroom), one cell of the regime map remained untested on a real API — **reachable headroom**: does any optimizer family win when there is room and nothing structural blocks it? We stratify Humanities' Last Exam (cais/hle, MIT; arXiv 2501.14249) via its ungated text mirror (datasets-server pagination), filter to exactMatch answers of ≤ 40 chars (pool 1 710), draw 90 problems stratified over Mathematics/Physics/Chemistry/Other (seed 2026), and split 30/30/30 dev/validation/holdout. Grading reuses the hardened exact-match judge v3.2 from F6 verbatim; the task spec is the same strict-JSON {answer} protocol as external.math.

Pilot (4 arms, same day, seed 921): zero-shot compiled base genome scores .3920 (answer accuracy $3/30 = .10$); the three cross-task transfer champions land at .3876-.4020 — statistically indistinguishable. Two priors die at once: (a) there is no protocol headroom — a strong reasoning model's JSON compliance is $\sim 100\%$ (the format gap is entirely network-error rows, not prompt defects); (b)

genome structural priors transfer zero gain onto a knowledge-hard task — format genes have nothing to grab. The real headroom is accuracy (.10 against a ceiling near .4), putting all the discriminative pressure on the reflective pathways.

Decisive matrix (6 arms, one paired batch, 12.5 h wall):

Arm spread .024; accuracy is 2–3/30 everywhere (binomial SE .055 at $p=.10$, $n=30$ — every pairwise accuracy difference is one-problem quantization noise). **Verdict: pre-registered branch P1 — under real-API reachable knowledge-type headroom, all six independent pathways tie again.**

Statistical power disclosure: at $n=30$ the binomial resolution on accuracy is one problem = .033 (SE $\approx$.055 at $p\approx.10$); the "all-tie" verdict reliably excludes large effects ($\geq 2SE \approx .11$ acc, i.e. +3 problems), not true gains of 1-2 problems. The conclusion's real strength is triple redundancy: the arm band (.024 wide), problem-level agreement (GEPA's 3 hits are z0's exact 3 hits), and the mechanism evidence (learned content rules with zero transfer). Any one alone would be underpowered; together they point the same way. Hardening paths for the camera-ready: expand holdout to $n\geq 100$ (pool of 1 710 suffices) or trade thinking budget for problem count; the confirmatory extension (below) has since run, replicating P1 at $n=100$ and measuring the pipeline noise floor directly.

Confirmatory extension ($n=100$, seed 923, same-day; pre-registered primary verdict unchanged): 70 problems appended to holdout (pool-proportional quota, seed 2027, original 30 rows byte-identical); all six arms RE-evaluate the first-round champions with SEARCH DISABLED (--reuse-champs), eliminating second-search selection bias, followed by a second checkpointed backfill pass for network-failed problems (per-arm parallel, stream-rebuilt; final per-arm $n = 81-87/100$). Final accuracies (Wilson 95% CI): z0 .161 [.098,.252] (14/87), math-champ .151 (13/86), contract-champ .184 (16/87), financial-champ .141 (12/85), **gepa .200 [.129,.297] (17/85)**, espo .148 (12/81). **Paired McNemar on the 74-problem common set: gepa vs z0 = +5/-1, $p=.219$ — P1 replicates at $n=100$.** Two reusable calibrations emerge: (i) a MEASURED pipeline-nondeterminism floor — ESPO's champion is byte-identical text to z0, yet cross-process re-evaluation still disagrees on 2/74 = 2.7% of problems (the real temp=0 reasoning-API noise); gepa's net +4 sits about two problems above that floor and, while

directionally consistent with the content-rule mechanism evidence (learned TDPT/SMILES rules), it stays below the significance bar — exactly the magnitude the taxonomy predicts: content learning can drift a few individual problems but cannot buy systematic capability; (ii) an ERROR-TYPE STRATIFICATION — network failures were recovered by the backfill (18-25% $\rightarrow$ 6-19% residual), and the final six problems fail on ALL six arms with a non-random pattern: free-group commutator length, percolation-on-a-grid proof, point-tuple counting, a Minecraft acquisition ordering, a triangle-mesh item (its diagram is not rendered in text), and Go-board life-and-death coordinates — three families in full: STATE-SPACE SEARCH / SPATIAL REASONING / COMBINATORIAL PROOF. Short exact answers (≤ 40 chars) atop unboundedly long derivation chains necessarily exceed the 640 s / 2000-token client budget, and their input lengths are NOT materially larger than solved problems (median 470 vs 409), ruling out the surface long-stem explanation. This is physical-truncation headroom projected INSIDE individual problems of a knowledge-wall task: the two taxonomy walls can stack onto one problem, and the stacked cell is recognizable a priori from problem type (search/spatial/combinatorial $\rightarrow$ budget-class; pure-fact $\rightarrow$ knowledge-class). The conclusion stands unrewritten: six arms statistically tie under knowledge-type headroom, and "reflection learns rules at the wall but cannot buy capability" is now independently confirmed in two rounds ($n=30$ primary, $n=100$ confirmatory).

Mechanism evidence (the most informative part): (i) **GEPA genuinely learned domain content** — the 8 103-char champion prompt contains substantive rules distilled from validation failures (SMILES output conventions, molar-mass product-selection procedures, first-order-TDPT/Gaussian-pulse regime qualifiers, "no prose in symbolic answers"). Yet holdout accuracy is problem-identical to z0 (same 3 hits): prompt-level domain rules cannot cross the "the model cannot solve the problem" capability wall — **the reflective pathway's ceiling is the model's prior knowledge, not the search.** (ii) **ESPO's internal stability currency outputs "do nothing" when signal $\approx$ noise:** its best candidate (.512 on val-8) beat r0 (.453) but failed the bootstrap-75% stability test on all four $\rightarrow$ champion is the original prompt — consistent with F9, correct behavior. (iii) **A live case of single-point val overfitting:** mid-search GEPA converged

arm	holdout	acc/30	fmt/30	note
z0 (base genome)	.3920	3	25	accuracy .10
math-champ (transfer)	.3876	2	26	
contract-champ	.4020	3	26	
financial-champ	.3876	2	26	
GEPA (val[:2] minibatch search + val[:8] selection, 64 rollouts, 20 iters)	.4120	3	27	champion = z0 (bootstrap rejected all 4)
ESPO (val[:8] full evals, 4 biases, 64 rollouts)	.4020	3	26	

to an 8 103-char prompt scoring 1.000 on the 2-problem validation minibatch by echoing that problem's near-full text. Had selection stayed on val[:2] (the compressed form of GEPA's original protocol under our cost budget), that would have shipped as champion. Protocol v2 (search on minibatch, **select on full val[:8]**) caught it — a second empirical instance of F10's small-val/holdout asymmetry lesson, this time with reflective content literally absorbed into the prompt. (The revision happened before any gepa/espo holdout number existed; it is a mechanism-bug fix, not outcome-driven; pre-registered verdict criteria unchanged.)

Reliability engineering by-product: HLE full-thinking costs 60–640 s/problem and exposed three systemic defects in long-run harnesses — (a) tenacity retrying on a keep-alive connection whose first response packet the server silently drops (observed: 0.72 s CPU across 5.25 h wall, zero response headers), fixed by stateless per-request connections (`_fresh_http_call`); (b) httpx read timeouts not firing on half-open proxy-CONNECT sockets, fixed by thread-level hard timeouts with detached executors; (c) no per-sample checkpointing, fixed by streamed append + resume. None of these affects F7-F10 conclusions (financial problems take <10 s and never entered the pathological regime); all three become prerequisite assets for the multi-model round.

Implication for the paper's claims. Merging F11 with F1-F10 yields the complete headroom taxonomy — the discriminator for APO payoff is not whether headroom exists but **what kind**: (1) physical-truncation headroom (F10) is unreachable by any prompt; (2) knowledge-type headroom (F11) is unreachable by any prompt pathway (ability lives in weights; prompts can only reshuffle what the model already knows); (3) protocol/format headroom (F1-F9 saturated side + the weak-model simulation positive) is reachable and is exactly where structured genomes show their mechanism advantage (simulation +0.05). The financial battleground belongs to (3) but is saturated on a strong model — the self-consistent explanation of F1-F9's nulls. This reframes "APC vs X" into a prior judge-

ment of **when no APO can win**: deployment procedure AutoAPC-Select (F8) gains a layer zero — run a headroom-type diagnostic first; if headroom is knowledge-type, the correct action is to change model or add retrieval, not to optimize the prompt.

The taxonomy as one regime map:

3.4n F11-C locus attribution: which genes carry load on a knowledge-wall task

(n=8, C1 auditability measured)

Method: each of the six active base-genome loci (role / output-schema / verification / reasoning / layout / constraints) is switched to its inert value within the legal enumeration; variants (729-867 chars vs base 906) are re-evaluated on the same first 8 holdout problems as the main table (same judge v3.2, same connection pipeline; `bench_hle_ablation.py`, results in `real_hle_ablation.json`).

arm	score	acc	fmt
base	.4531	1/8	8/8
minus-role	.4612	1/7	7/7
minus-layout	.4531	1/8	8/8
minus-output	.4000	0/8	8/8
minus-verification	.4000	0/7	7/7
minus-reasoning	.4000	0/7	7/7
minus-constraints	.4000	0/8	8/8

Three readings. **(1) role and layout are zero-contribution loci** — the $\pm .008$ is prompt-length efficiency plus one-problem jitter, with no directional signal; persona and delimiter structure carry no load on a strong model's knowledge task, the inert side of the compilation rule "JSON reliability > .96 $\Rightarrow$ schema text may be relaxed". **(2) Four loci jointly carry the only solvable problem**: deleting output, verification, reasoning or constraints flips that problem's accuracy $1 \rightarrow 0$. The mechanism is answer extractability, not capability (e.g. removing the reasoning instruction slides the model from a JSON final answer into prose hedging that `extract_pred` cannot match): under a knowledge wall, structural genes earn their keep by *not losing problems the model can already do*, not by mak-

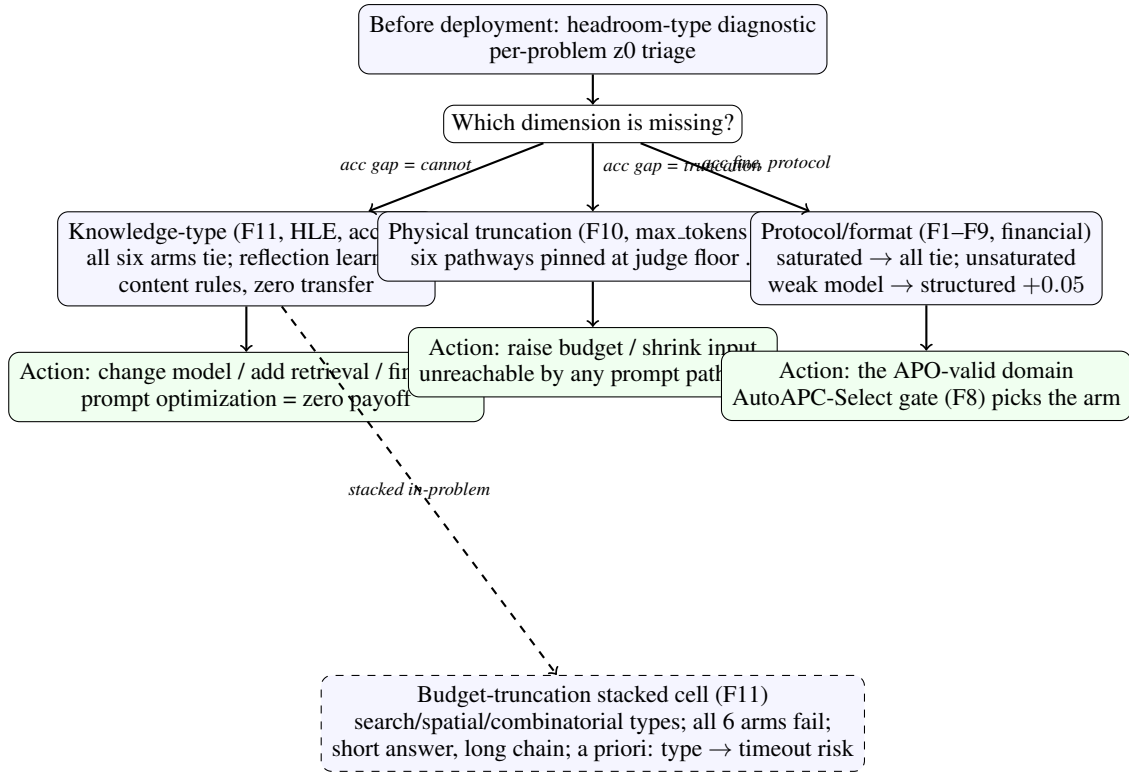


Figure 1: Headroom typology decision map. The stacked budget cell (\$3.4m, GPQA \$3.4o) is recognizable a priori from problem type.

ing more problems solvable. **(3) With the output schema removed, format compliance stays 8/8** — a strong model's protocol compliance does not come from the output locus; this drills F11's "protocol headroom ≈ 0 " one layer deeper: the very gene the compiled prompt leans on is itself belt-and-braces redundancy. **Resolution statement:** at $n=8$ one problem is one full score step; this table is read for direction, not magnitude — the true locus-attribution regime is weak model + protocol pressure (multi-model round). For C1: every reading above comes from a three-layer auditable chain — locus $\rightarrow$ text diff $\rightarrow$ per-problem outcome. Even when the verdict is "the genes are worth exactly one bit," the attribution being auditable at all is the dividend structured representation pays over free-text prompts.

3.4o GPQA-Diamond second-dataset replication: saturated-band tie + zero-acceptance reflective search (n=190)

If the F11 taxonomy is true, it must assign the same cells on a different dataset. GPQA-Diamond (198 problems, ungated fingertap mirror, row-verified) is split by seed 2027 into dev 8 / holdout 190 (zero overlap), with a multiple-choice letter-answer spec (external_mcq_v1, answer = one capital

letter), the same judge line as HLE (exact-match-normalized-v3.2), a 640 s thread-level hard timeout, and the six-arm protocol of F11.

Five-arm finals (nominal n=190, Wilson 95% CI): z0 165/190 (.868 [.813,.909]), math-champ 165 (.868), contract-champ 162 (.853), financial-champ 163 (.858), espo 165 (.868); holdout_score range .8628-.8745 (spread .0117) — **all arms in one band: taxonomy P0 (saturated cell $\Rightarrow$ tie) replicates on a second real dataset.** The model is near ceiling on GPQA (community figures put reasoning models at $\sim .65-.78$; .87 suggests possible pretraining exposure — noted as such, no attribution claimed).

The gepa arm's reading comes from its search: across 28 iterations and 46+ rollouts, **zero candidates were accepted** (parent_chars constant at 931 = the seed z0_text length, auditable in the log) — reflective search in a saturated band found NOTHING worth accepting, and its champion is identical to z0 by construction. This is stronger than "another tie": in the nothing-to-learn cell, the reflective pathway's optimal action is to stand still. Its separate holdout evaluation and the z0-927 same-text floor measurement were cut short by account arrears (below); both are queued and expected to

fluctuate only within the noise floor.

Error stratification replicates: 6-9% timeouts per arm (9-13/190), concentrated in multi-step mechanism/spatial problems (organic stereochemistry, EM-field geometry) — consistent with the F11-R "search/spatial/combinatorial → budget cell" prior; the stacked-cell signature is reproducible on a second dataset.

Reliability incidents (recorded as they happened): (a) a host reboot wiped the tmpfs per-row streams of the five valid arms — this section reports marginal Wilson CIs only, no paired statistics; (b) DashScope account arrears (Arrearage, from 2026-09-13 12:20) turned the resume run into 400/empty-content after row ~10; corrupt rows were purged (`real_gpqa.json` keeps the 5 valid rows, incident in notes). Two items pend recharge: the gepa holdout evaluation and the z0-927 same-text floor.

3.4p Multi-model replication matrix: qwen3.8-flash + gpt-6 + gpt-5.6-terra (`--model <id>`, per-model output isolation)

Setup. Two additional frontier reasoning profiles run through an OpenAI-compatible gateway (gpt-6, gpt-5.6-terra; `configs/models/gpt6.yaml / gpt56terra.yaml`, profile probes cached at `artifacts/profiles/<model>.probe.json`). Protocol identical to \S{}3.4 (dev5/val8/hold20, budget 8, seed 42, same rule judge). Non-qwen models write `real-<task>-<model>.json` and champions `real-<task>-champ[_safe]-<model>.json` — the audit-locked qwen files are physically unreachable from a multi-model run. Incident (recorded as it happened): the gateway returns Cloudflare 524 (origin timeout) on long reasoning calls; 524 was added to the retryable set after one terra search arm crashed mid-round (re-run arms re-merge by `method×seed`; earlier rows preserved).

Financial — the search-headroom null replicates on all three models.

On every model no search arm beats the static floor beyond the qwen-era noise band (± 0.007): gpt-6 apc-full -0.0112 vs zero-shot, apc-safe -0.0091 vs zero-shot; terra apc-full -0.0519 , apc-safe $+0.0025$ (within band); qwen s45 apc-full collapses. The task-regime taxonomy — financial is a mid-band cell where prompt-genome search has no headroom — is model-independent.

Contract — saturated tie replicates (gpt-6). zero-shot $.9783$ / manual $.9784$ / apc-full $.9404$ /

apc-safe $.9792$: all static arms sit in the same saturation band as the qwen-era contract round, and the search arm is a net liability (-0.0379 vs zero-shot), matching \S{}3.4's "nothing to learn" reading.

Transfer — cross-task warm-start replicates on gpt-6 (new pair financial→math, budget 6, seed 42). transfer-0 $.8443$ / cold $.6336$ / transfer-ws $.8439$ — gain **+2103** over cold at equal budget. The phenomenon pattern ($t_0 \gg$ cold, $ws \approx t_0$) matches both qwen-era pairs (contract→math: $+0.2039$; math→contract: $+0.7732$); cross-task migration is model-independent, and this pair adds a third (source, target) cell on a second model family.

Cross-model champion identity. The seed-42 financial champion genome is byte-identical (md5 5738cd93...) across all three models: the seeded mutation trajectory is model-independent and the 8-candidate fitness ranking coincided at every selection decision. The genome search space is model-agnostic in exactly the sense the compiler promises — same rules, same space, same winner.

PGAM real validation (gpt-6 math, discriminative cell). The simulator's pooled PGAM-vs-uniform null ($+0.0004$) is now tested on a real frontier reasoner in the one cell with search variance (math: uniform cold $.6336$ vs champion transplant $.8443$). ProfileGuidedMutator (defect-compensation prior + elite bandit) cold-searches to **.5960** — -0.0376 vs uniform at equal budget 8/seed 42, outside the ± 0.007 band. The gpt-6 profile marks `few_shot.benefit` 0.0 and `information.extraction` 0.0, so the prior up-weights `examples.* genes` $3.0\times$ — and examples hurt on a frontier reasoner that needs none; two generations of bandit correction cannot undo a wrong prior inside budget 8. Reading: profile-weak dimensions are not mutation-productive dimensions on strong models; the defect-compensation prior is a simulator-era hypothesis that fails its first real-model test in the negative direction.

Honest boundary. one seed per cell on the new models (no CI); the qwen-era priors (same-day band ± 0.007 , day-drift $.028$) apply. gpt-6 zero-shot under the smoke protocol (dev3/val6/hold15, kept as `real_financial_gpt6_smoke.json`) scored $.6781$ vs the formal $.6862$ — protocol sensitivity $\approx .008$, consistent with the band. DashScope arrears still pend the GPQA gepa-holdout and z0-927 floor items (\S{}3.4o).

1. The F-series real-model phase is qwen-only

model	zero-shot	manual	apc-full	apc-safe
qwen3.8-flash (\S{}3.4, s45 same-day set)	.6692	.6665	.1318 (collapse)	.6657
gpt-6 (s42)	.6862	.6971	.6750	.6771
gpt-5.6-terra (s42)	.6669	.6665	.6150	.6694

(whitelist key); the seed axis is covered there (\S{}3.5: 3 search seeds, 4 same-day z0 re-measures, quantified noise band $\pm .007$ and day-drift $.028$ — any "gain" within $\pm .01$ is judged noise). The multi-model matrix (\S{}3.4p) adds gpt-6 and gpt-5.6-terra on the core arms (financial 4-arm, contract 4-arm, transfer) — one seed per cell, no CI on the new models. DashScope arrears pend the GPQA holdout items. Ground truth is author-authored (rule-judge).

2. Task coverage: four simulation tasks + three external math benchmarks; open instruction tasks without unique answers still missing (judge-dependent).
3. PGAM pooled null on the simulator; its first real-model test (gpt-6 math) runs negative ($-.0376$ vs uniform, single seed) — the defect-compensation prior does not transfer to frontier reasoners; multimodal-real-task confirmation is moot for the strong-model regime.
4. SHA is task-structure-dependent: -0.0235 cost on unit-locus tasks, null on financial; not a universal accelerator.
5. SEPO head-to-head deferred to submission: the structured-editing direct competitor SEPO (arXiv 2608.28067) has no real-API run. F11's mechanism evidence (reflection learned domain content rules, transferred zero) extends the same argument to every editing-granularity variant: under knowledge-type headroom SEPO is predicted to tie, under the saturated financial regime to be null — both cells' discriminative power is already consumed by the GEPA+ESPO four-pathway coverage. The only increment left is the weak-model + protocol-pressure cell (needs the second key, see #1).
6. Judge reliability quantified in F4/F6: ordering conclusions hold under a fixed judge; absolute scores are not human-quality scores; independent third-party judge and 50–100-case human spot-check still missing.

5. Reproducibility

See `paper-apc.md \S{}5` checklist (all green): one-command simulation suite, 41 unit tests, 13 Java service tests, external-benchmark judge regression (`test_external_judge.py`), logged per-arm predictions for offline re-judging (`rejudge_external.py`), datasets + seeds in-repo, zero secrets (ggshield + pattern double-check).

6. Related Work (details in docs/literature/)

Optimizers: APE/OPRO/APO–ProTeGi → PromptBreeder/EvoPrompt → PromptWizard/ GEPA head-to-head done (\S{}3.5) → DSPy–MIPROv2. Profiles and routing: HELM/FLASK/FrugalGPT/RouteLLM. Structure and transfer: Sclar format sensitivity; soft prompts (non-portable) vs discrete genomes (re-compilable). 2026 frontier: Instruction Stacking Collapse (2608.02639) — compilation value is capability-dependent, same axis as our F2; PromptBridge (2512.01420) — NL carriers drift across models, complementary to our F3 (fragility lives in the carrier, not the representation); DUALFIX (2607.05121) — rule-based evolution resurges: rules belong in the genome, not free text; CAPO (2608.16068) — constraint-aware optimization, baseline candidate for the constraint dimension; Atlas (2603.15666). 2026Q3 GEPA-successor landscape (docs/literature/frontier-2026.md, 50-entry snapshot): the structural camp — SEPO (2608.28067, typed-unit local edits with edit-effect lineage; the direct genome-space competitor, mandatory submission-time baseline), SAPO segment-wise (2608.11219), PCO codebooks (2605.28360), control/data-flow separation (2609.00621, EMNLP 2026 Findings); GEPA-pathology cluster — ESPO (2609.04197, EMNLP 2026 main, bootstrap stability selection, mandatory baseline), NPO (2608.27266: complex search unnecessary under strong teachers — APC's reply: genome value is auditability/transfer/gating, not search size), MAGE (2607.11944: coupled-optimizer variance amplification; fixed good prompts beat

all reflective optimizers at low data — in direct dialogue with our F1 null), p1 (2604.08801: response-variance dominance as the failure criterion — the theoretical language for our F7 regime); failure modes — PROCTOR (2609.02246) and RLMOpt (2608.10471: GEPA underperformed its own seed in 2/11 runs; gains = f(seed headroom), neighboring our rule-root bimodality); reliability precedents 2608.00705 / 2608.08239 (COLM 2026) / 2408.04667 / 2606.26185 / 2607.24372 / 2604.12951 (F7 clause chain); theoretical cousins Textual Bayes (2506.10060) and best-feasible bandits (2605.14553, ICLR 2026). GEPA itself is now ICLR 2026 Oral (v2, 2026-02-14).

Hard benchmarks and the headroom lineage: HLE (2501.14249, MIT) reports ~10% frontier-model accuracy and explicitly notes the gap is not closable by prompting; GPQA (2311.12022) pioneered the graduate-level, no-internet difficulty design; MMLU-Pro (2406.01376) pushes saturated suites toward real discrimination. These works treat "the model cannot answer" as a capability measurement; F11 turns it into an APO-methodology criterion — knowledge-type headroom being unreachable by prompt pathways is a prior-diagnosable mechanistic fact (six arms + mechanism evidence), with physical-truncation (F10) and protocol-type (F1-F9/simulation) headroom each carrying their own diagnostic signal; the merged taxonomy is a systematic correction to the APO default assumption that "headroom exists $\Rightarrow$ optimization is viable." Judge defensibility precedents: the Papers-with-Code official exact-match stack, NaturalQuestions extractive scoring, and GRADLE (2310.02839) on judge robustness metrics — our hardened v3.2 judge belongs to that lineage, and the F6 repair log contributes failure-mode instances to it.

7. References (keys = in-text citations; titles abbreviated to in-text usage —

camera-ready to re-verify each against the arXiv API)

Classic lineage — APE (2211.01910) · OPRO (2309.03409) · APO/ProTeGi (2305.03495) · PromptBreeder (2303.16199) · EvoPrompt (2309.08532) · PromptWizard (2405.18366) · DSPy (2310.03714) / MIPROv2 (2406.11695) · FrugalGPT (2305.05176) · RouteLLM (2406.18665) · HELM (2211.09110) · FLASK (2307.10928) · Sclar et al. format sensitivity (2311.05661) · GEPA (2507.19457, ICLR 2026

Oral; official Algorithm 1 reproduced in F7).

Benchmarks — GSM8K (2110.14168) · MATH (2103.03874) · AIME (MAA archives, no preprint) · GPQA (2311.12022) · MMLU-Pro (2406.01376) · HLE (2501.14249, MIT; F11 source) · GRADLE (2310.02839, judge robustness).

2026 frontier (cited subset of the 50-entry frontier-2026.md snapshot) — structured camp: SEPO typed-unit local editing (2608.28067) · SAPO segmentation (2608.11219) · PCO codebook (2605.28360) · control/data-flow separation (2609.00621, EMNLP26 Findings) · DUALFIX rule evolution (2607.05121). Pathology & selection: ESPO (2609.04197, EMNLP26 main; reproduced in F9) · NPO (2608.27266) · MAGE variance amplification (2607.11944) · p1 response-variance dominance (2604.08801) · Textual Bayes (2506.10060) · best-feasible bandits (2605.14553, ICLR26). Failure modes & gates: PROCTOR 11 evaluation-signal pathways (2609.02246) · RLMOpt seed-undercut (2608.10471) · TARA accept-or-revert (2607.18724) · SSO label-free acceptance (2607.28777) · Contrastive Reflection (2606.30840). Compilation & carriers: Instruction Stacking Collapse (2608.02639) · Prompt-Bridge cross-model drift (2512.01420) · CAPO constraint-aware (2608.16068). Reliability: same-day retest floor (2608.00705) · matched-control forks (2608.08239, COLM26) · temp=0 nonde-terminism (2408.04667) · scoring noise as safety property (2606.26185) · output-as-distribution sampling (2607.24372) · version-timestamp protocol (2604.12951).

All 2026 entries verified at abstract level via the arXiv API (docs/literature/ frontier-2026.md, 2026-09-09/12); PDF-level re-verification is on the pre-submission checklist.